PROJECT DOCUMENTATION

AI Career Roadmap Generator

PathPilot

A responsive web application for role-based skill-gap analysis and structured career learning plans

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GitHub Repository: https://github.com/blessypriya2004-design/AI-Career-Roadmap-Generator

Deployment URL (provided by team): https://ai-career-roadmap-generator-8fau.onrender.com

Prepared from the public project README and source files. See the implementation notes and limitations before final academic submission.

Declaration and Project Snapshot

This document describes PathPilot, a web-based career planning prototype. It is intended to help a learner compare their currently listed skills with a predefined skill checklist for a selected technology role and receive a structured six-week learning plan.

Project status note: The repository describes the current implementation as rule-based. Although the project title contains the word “AI”, the current starter version does not call an external AI model or train a machine-learning model. This distinction is stated clearly to avoid overstating the implementation.

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1. Abstract

PathPilot is a responsive career planning application built using HTML, CSS, JavaScript, Python and Flask. The user provides education, a target role, experience, current skills and weekly study hours. The application compares the supplied skills against role-specific requirements, identifies missing skills, calculates a checklist match percentage, and generates a six-week learning plan containing weekly focus areas and practical tasks. The user can download the generated roadmap as a text file. The current implementation uses predefined rules and role data rather than a generative AI service.

2. Introduction and Background

Technology careers require a combination of technical knowledge, practice and project experience. Learners may find it difficult to decide which skills to study first or how to organise their preparation. PathPilot aims to make this planning process more structured by mapping a selected career role to a list of required skills and a short sequence of weekly learning activities.

The application is designed as a lightweight web prototype. A browser-based interface collects learner inputs, JavaScript communicates with a Flask backend, and the backend returns roadmap information in JSON format.

3. Problem Statement

Learners often have an incomplete understanding of the skills expected for a target technology role. Without a clear checklist and practical sequence, learning can become unstructured. The project addresses this by providing a role-based skill checklist, highlighting missing skills and suggesting a six-week plan with practical tasks.

4. Objectives and Scope

- Collect basic learner information and a chosen target technology role.
- Compare user-entered skills with predefined role requirements.
- Display matched skills and skill gaps.
- Calculate a role checklist match percentage.
- Generate a six-week learning plan with focus topics and practice tasks.
- Provide recommendations and a text-file download option.
- Offer a responsive interface accessible from desktop and mobile browsers.

Scope: The current prototype supports four predefined roles: Python Full Stack Developer, Frontend Developer, Data Analyst and Machine Learning Engineer. It is a planning aid, not a recruitment, certification or job-readiness assessment tool.

5. Existing and Proposed System

Existing approach

A learner may search multiple websites, job descriptions and tutorials manually, then create a personal checklist and timetable. This can require considerable effort and may lead to an inconsistent learning sequence.

Proposed approach

PathPilot consolidates a basic role checklist and a six-week learning outline in one interface. It automates the comparison of entered skills against predefined requirements and presents the gaps and suggested tasks in a consistent format.

6. Requirements and Technology Stack

Software requirements

- Python 3.10 or newer (as stated in the repository README).
- Flask web framework.
- A modern web browser.
- Git for source-code version control.
- pytest for running the automated tests.

Technology stack

Layer	Technology	Purpose
Frontend	HTML5, CSS3	Page structure, layout and responsive styling
Client logic	Vanilla JavaScript	Input handling and API requests
Backend	Python, Flask	Routes, validation and roadmap generation
Data	Predefined Python dictionaries	Role requirements and weekly plan content
Testing	pytest	Automated application tests
Deployment	Render, Gunicorn	Hosted Flask web service

7. System Architecture

The application follows a simple client-server architecture. The browser renders the user interface and sends JSON requests to the Flask backend. The backend validates the request, executes the rule-based roadmap function and returns a JSON response. Static frontend assets are served by Flask.

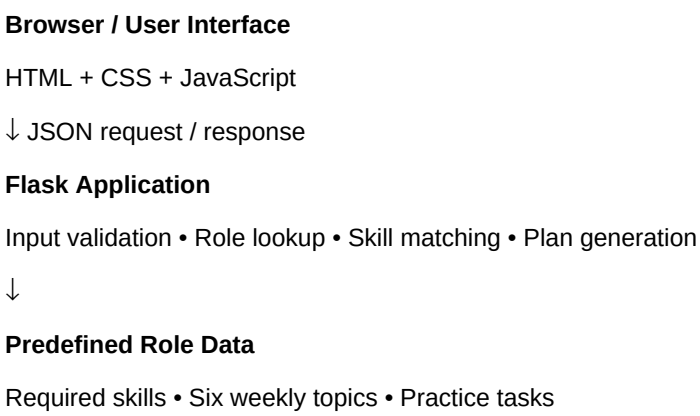

Figure 1. High-level logical architecture (conceptual).

8. Module Description

8.1 User Input Module

Collects education, target role, experience, current skills and weekly study hours from the user interface.

8.2 Role Selection Module

Provides four supported technology roles. Each role is associated with a predefined list of required skills and six weekly learning units.

8.3 Skill Normalisation and Matching Module

Accepts a comma-separated skill string or a list of skills. Common aliases such as “js” for JavaScript, “html5” for HTML, “mysql” for SQL and “ml” for Machine Learning are normalised before comparison.

8.4 Skill Gap Analysis Module

Separates required skills into matched skills and missing skills. The checklist percentage is calculated as matched required skills divided by the total required skills, multiplied by 100 and rounded to the nearest whole number.

8.5 Roadmap Generation Module

Creates six weekly entries for the selected role. Each entry includes a title, focus description, practical task and the learner's selected weekly study-hour value.

8.6 Recommendation and Download Module

Returns recommendations to begin with a missing skill, practise through projects, document work in GitHub and compare the plan with real job descriptions. The frontend also provides a text-file download option.

9. Data Flow and API Design

Typical data flow

- The learner opens the PathPilot web page.
- The learner enters education, target role, experience, skills and weekly study hours.
- The frontend sends the input to the Flask API as JSON.
- The backend validates the target role, education and study hours.
- The backend normalises the skills and compares them with the selected role requirements.
- The backend prepares matched skills, gaps, checklist percentage, six weekly tasks and recommendations.
- The frontend displays the returned roadmap and allows a text download.

API endpoints

Method	Endpoint	Purpose
GET	/api/roles	Returns the list of supported roles.
POST	/api/roadmap	Accepts learner input and returns a generated roadmap.
GET	/	Serves the main web page.

POST /api/roadmap input fields

Field	Description
education	Learner's education; required.
target_role	One of the four supported roles; required.
experience	Experience description; defaults to “Fresher” if omitted.
skills	Comma-separated string or list of current skills.
hours_per_week	Integer from 1 to 40; defaults to 7 if omitted.

The API returns the target role, education, experience, matched skills, skill gaps, match percentage, roadmap entries, recommendations and a note describing the rule-based nature of the current version.

10. Algorithm and Roadmap Generation

- Read the target role and verify that it exists in the predefined role data.
- Check that education is not empty.
- Convert weekly study hours to an integer and validate the range 1–40.
- Read the skills input and split a comma-separated string when necessary.
- Normalise known aliases and compare the learner's skills with the role's required skills.
- Create matched and missing-skill lists.
- Compute the checklist match percentage: (number of matched skills / number of required skills) × 100.
- Attach the predefined six-week plan for the selected role.
- Return recommendations and the generated result as JSON.

11. Installation and Execution

The repository README documents a Windows-based local setup. Use the project root directory in the terminal.

1. Create a virtual environment

```
py -m venv .venv
```

2. Activate it

```
.venv\\Scripts\\activate
```

3. Install dependencies

```
pip install -r backend/requirements.txt
```

4. Start the backend

```
python backend/app.py
```

5. Open in a browser

```
http://127.0.0.1:5000
```

Deployment note: The team has been working on a Render deployment using Gunicorn. The exact Render start command, application entry point and static-file directory must match the final committed repository structure. Confirm the deployed home page and API before treating the deployment URL as verified.

12. Testing

The repository includes a basic automated test file and documents pytest as the test runner. The following test cases are recommended for demonstration and evaluation. Record actual pass/fail results after running the current code; the table below is a test plan, not a claim that every test has passed.

Test ID	Test scenario	Expected result
TC-01	Request supported roles	API returns the four configured roles.
TC-02	Submit valid learner details	Roadmap JSON is returned.
TC-03	Submit unsupported target role	Validation error is returned.
TC-04	Leave education empty	Validation error is returned.
TC-05	Enter study hours below 1 or above 40	Validation error is returned.
TC-06	Enter known skill aliases	Aliases are normalised before matching.
TC-07	Enter all required skills	No skill gaps are reported and the checklist reaches 100%.
TC-08	Open the home page and load assets	HTML, CSS and JavaScript load successfully.

Run automated tests using:

```
pip install pytest
pytest -q
```

13. Results and Expected Output

For a valid request, the application is expected to display the selected target role, the learner's matched skills, missing skills, a checklist percentage, six weekly learning units and recommendations. The roadmap can be downloaded as a text file. The exact output depends on the user's selected role and entered skills.

The percentage represents only the proportion of predefined checklist skills matched. It should not be interpreted as a validated employability score, proficiency assessment or prediction of hiring success.

14. Limitations

- The current starter version uses predefined rules and role data; it does not call an external AI model or train a machine-learning model.
- Only four target roles are included in the predefined role data.
- The skill checklist is simplified and may not cover every employer's requirements or emerging technologies.
- The application does not currently provide user accounts, persistent profile storage or saved learning history.
- The checklist percentage is not a measure of job readiness and does not guarantee employment.
- Roadmap recommendations are general guidance and should be compared with current job descriptions and learning resources.

15. Future Enhancements

- Integrate a verified AI service to generate more adaptive roadmaps, with clear disclosure and safe handling of API credentials.
- Add more career roles and allow administrators to maintain role-specific skill data.
- Add user registration, secure login and database-backed profiles.

- Allow users to track weekly progress and revise their learning plan.
- Add curated learning resources, project links and interview practice.
- Improve validation, accessibility, automated test coverage and deployment monitoring.

16. Conclusion

PathPilot provides a simple and structured starting point for technology career planning. By combining role-based skill checklists, skill-gap identification and a six-week learning outline, it helps learners organise their next learning steps. The current version is a rule-based prototype. Future work can expand the role catalogue, introduce progress tracking and, if required, add a carefully integrated AI service.

17. References

- Project repository and README: https://github.com/blessypriya2004-design/AI-Career-Roadmap-Generator
- Flask documentation: https://flask.palletsprojects.com/
- Python documentation: https://docs.python.org/3/
- pytest documentation: https://docs.pytest.org/
- Render documentation: https://render.com/docs

— End of Documentation —